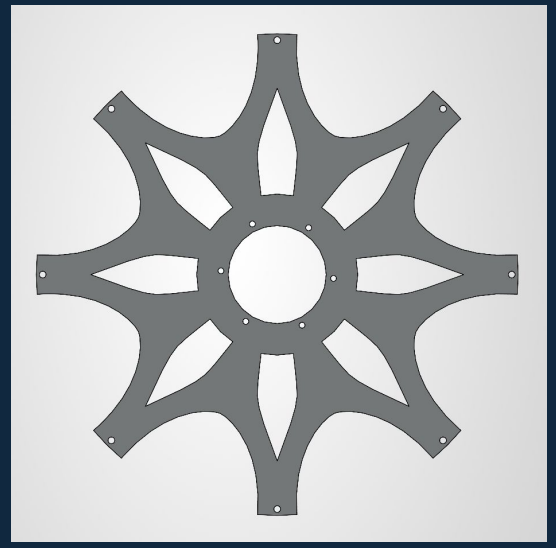
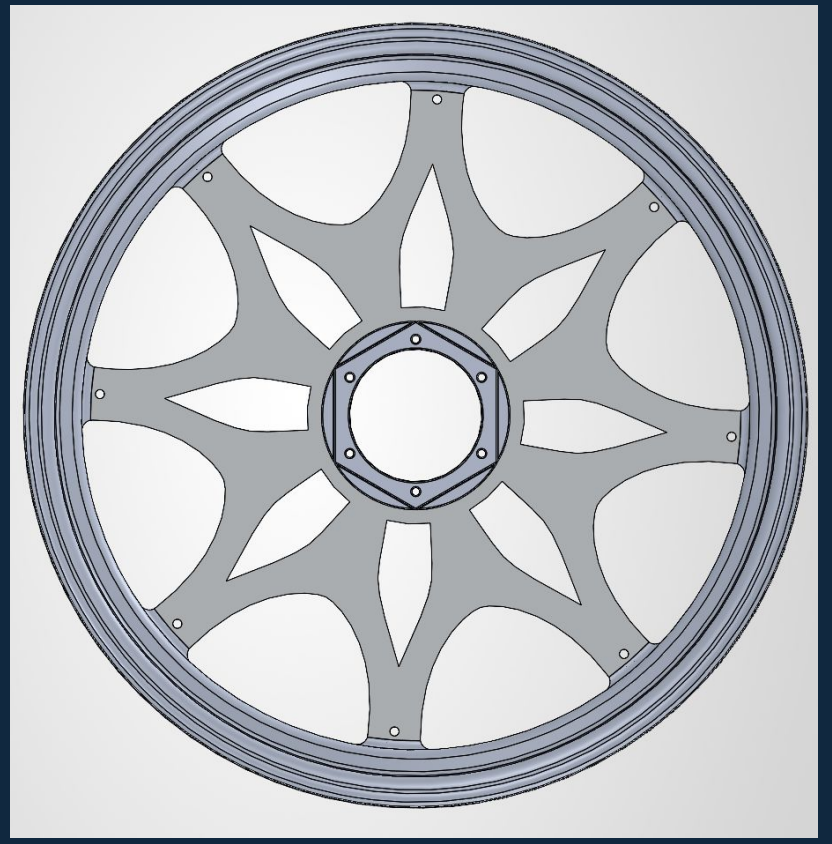
Wheel Spoke Design

CalSol Dynamics onboarding / Fall 2026

Vincent Ly, Sean Chen





Design objective and constraints

A lighter spoke assembly that fits the supplied rim and centerlock

Required load cases

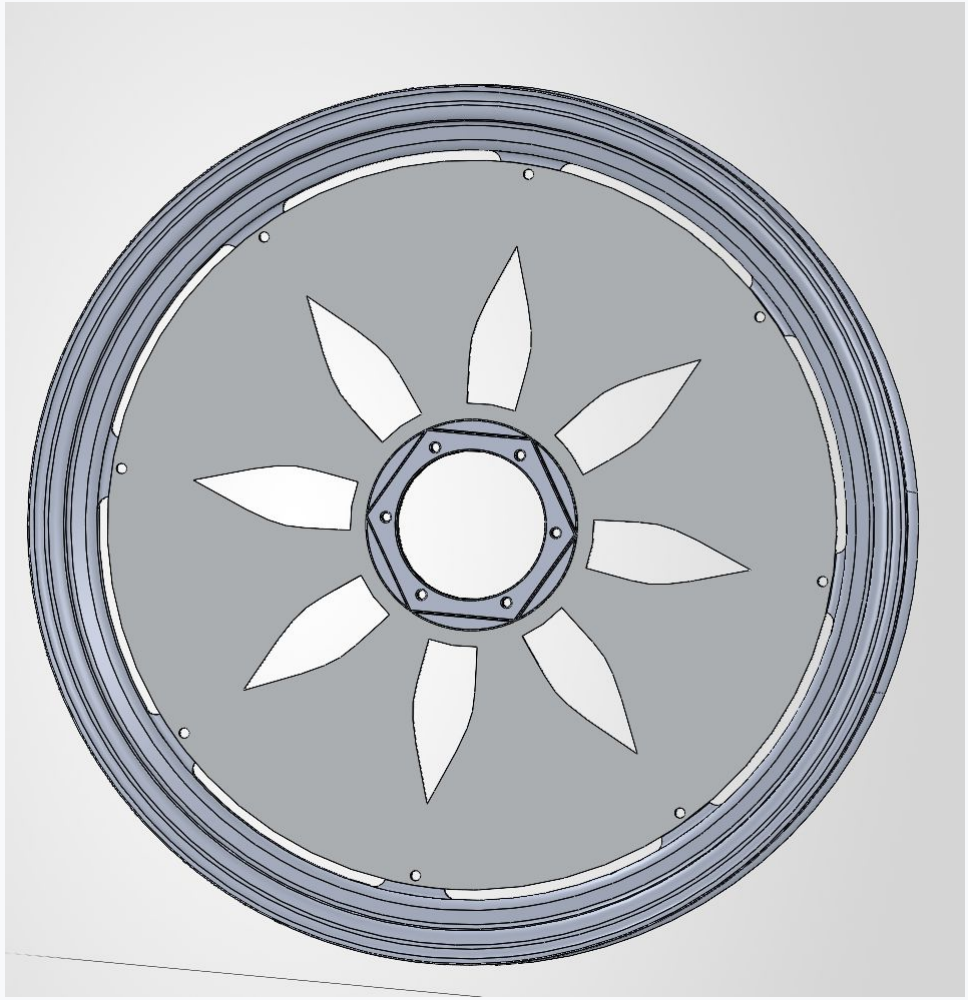
Separate static studies: 2g bump and 1g braking.

Assembly interfaces

- Connected with pins
- Centerlock
- Rim

Design target

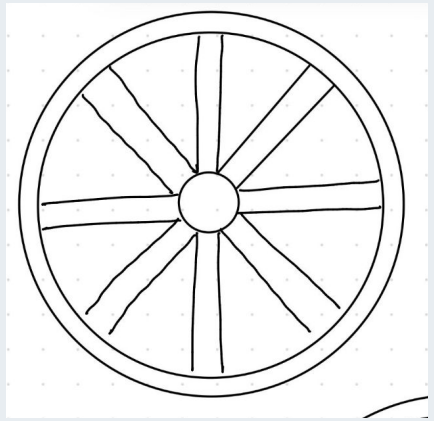
FOS above 1 in both cases. Compare spoke mass against 698 g.



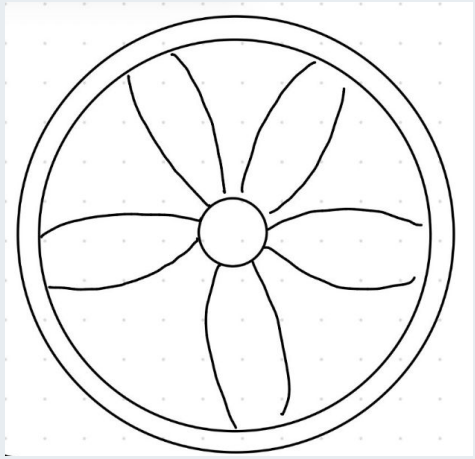
Concepts and design selection

Show how your initial sketches led to the modeled design

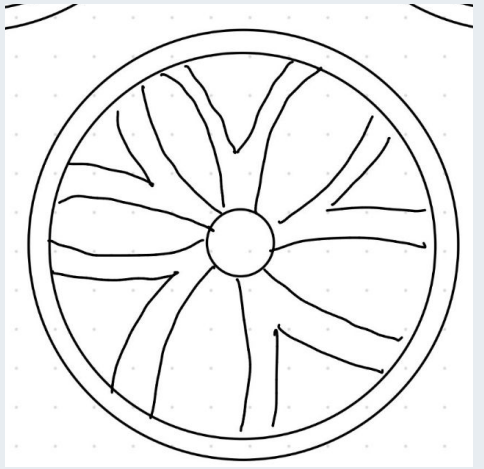
First Sketch



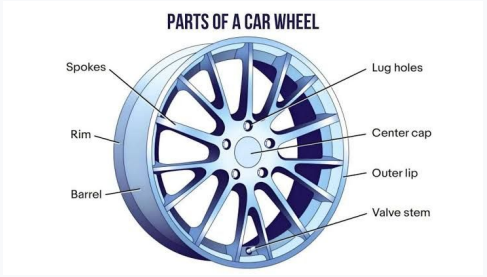
Second Sketch



Final Sketch



- Based on online references
- Get the load from the rim to the center
- Distribute the load evenly





CAD geometry, assembly, and material

Spoke geometry

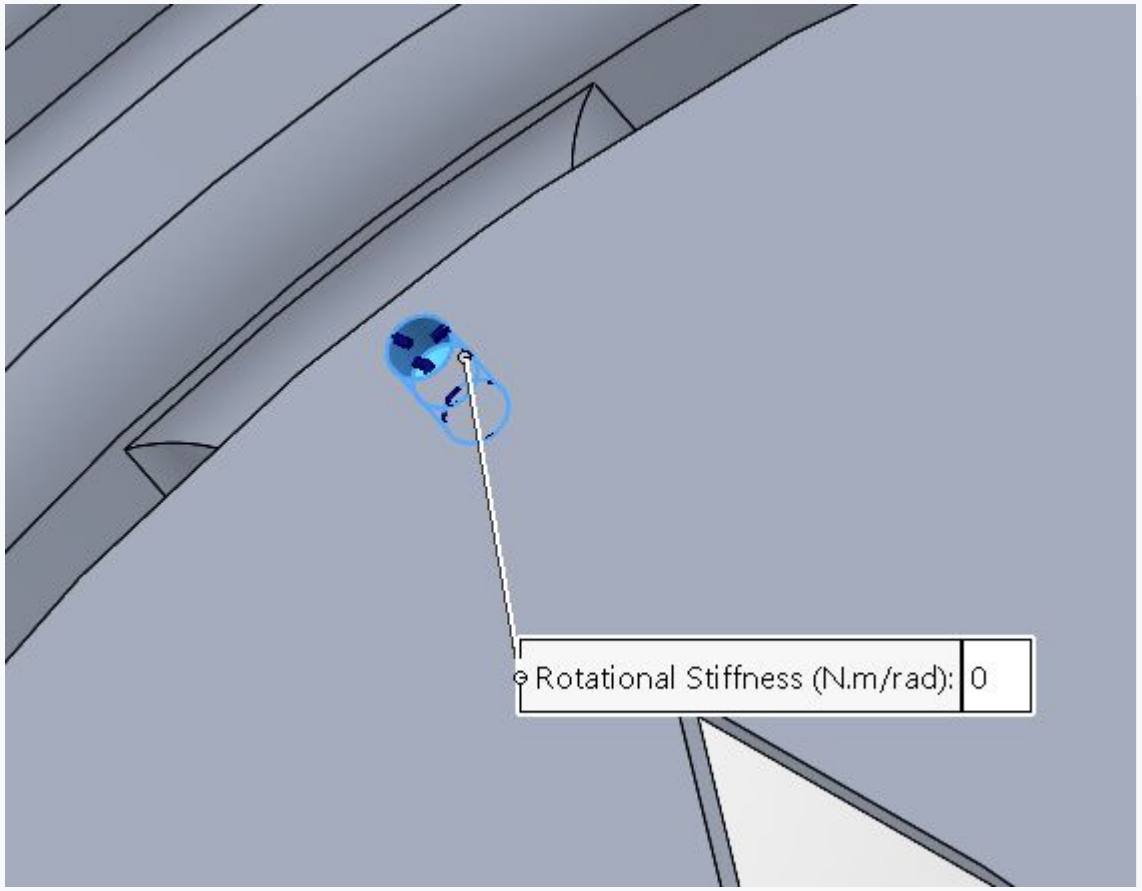
Thickness: 5 mm
8 Spokes

Mounting features

- Used pins in FEA to connect
- Mounting points on rim and center

Modeling decisions

- Wanted to distribute the loads better so we added more paths for the loads to go down
- Also wanted to save weight so we cut out holes
- Smooth to reduce stress concentrations



Load calculations and assumptions

Show the calculation behind each study

BUMP / 2g

$F = (318\ kg / 4) \times (2 \times 9.8\ m/s^2)$
$F = 1{,}558.2\ N$

Assumption: equal weight on four wheels.

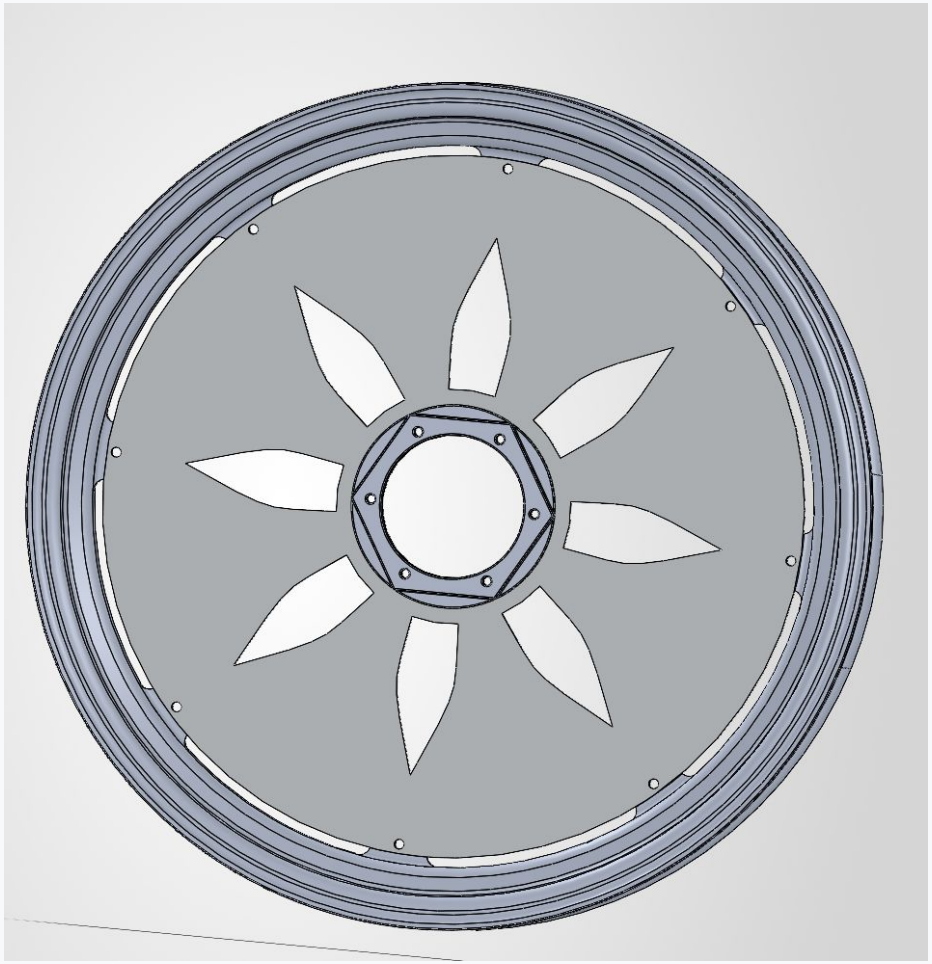
$\Sigma F_y = 4F_{Wheel} - W_{car} = 0$

BRAKING / 1g

$F = (318\ kg / 4) \times 9.8\ m/s^2$
$T = F \times R = 168.79\ \ N \cdot m$

$R = 0.216\ m$

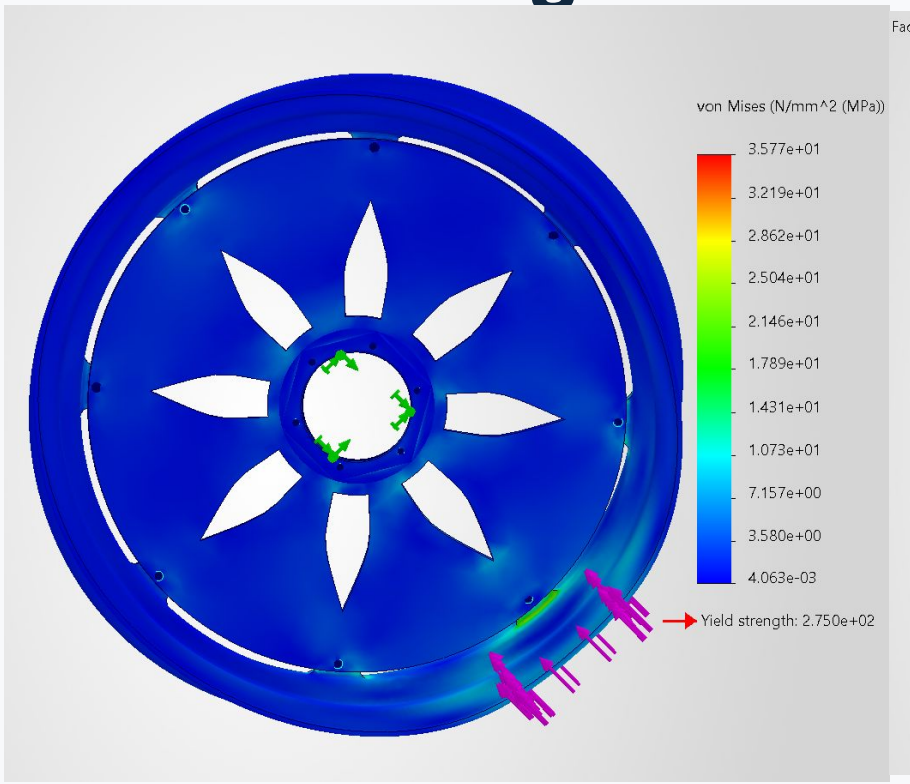
First Design

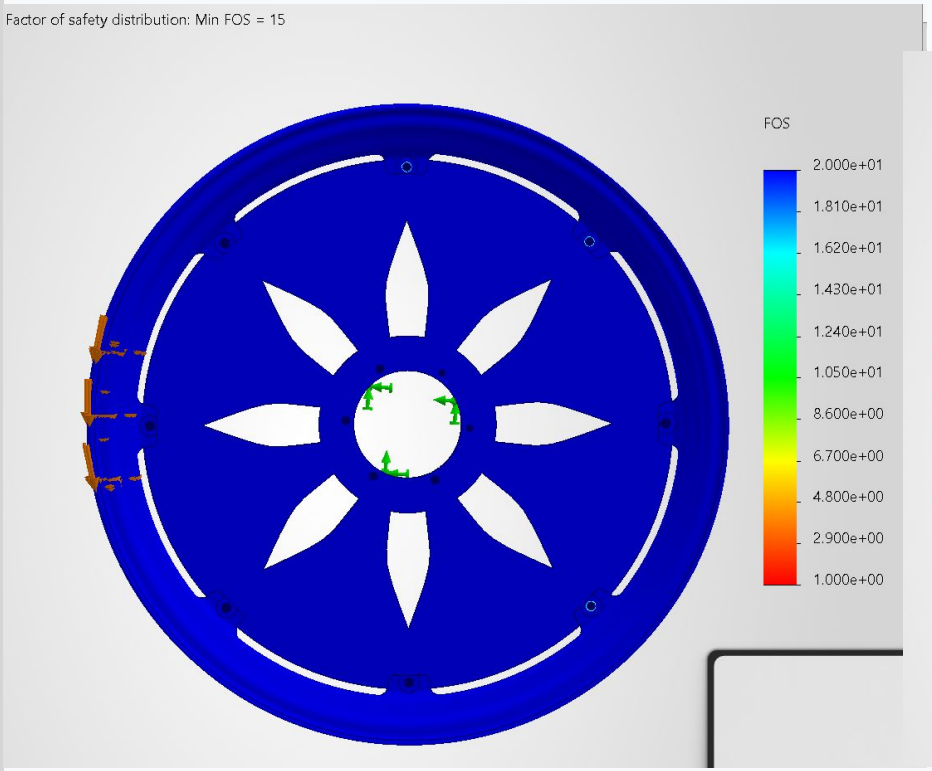


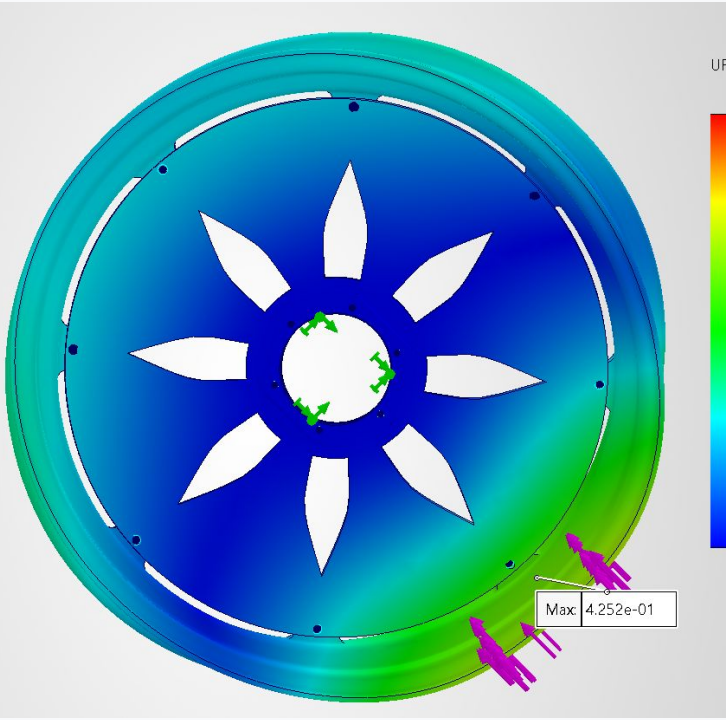
Decision:

- Strong, can resist lots of forces
- High FOS
- 6061-T6

First Design



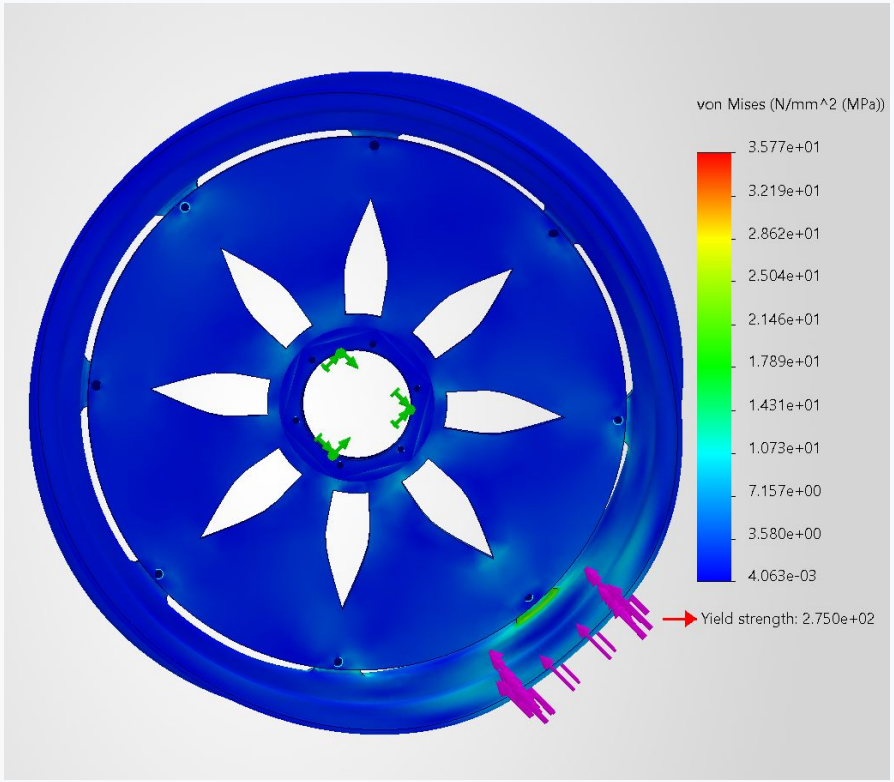


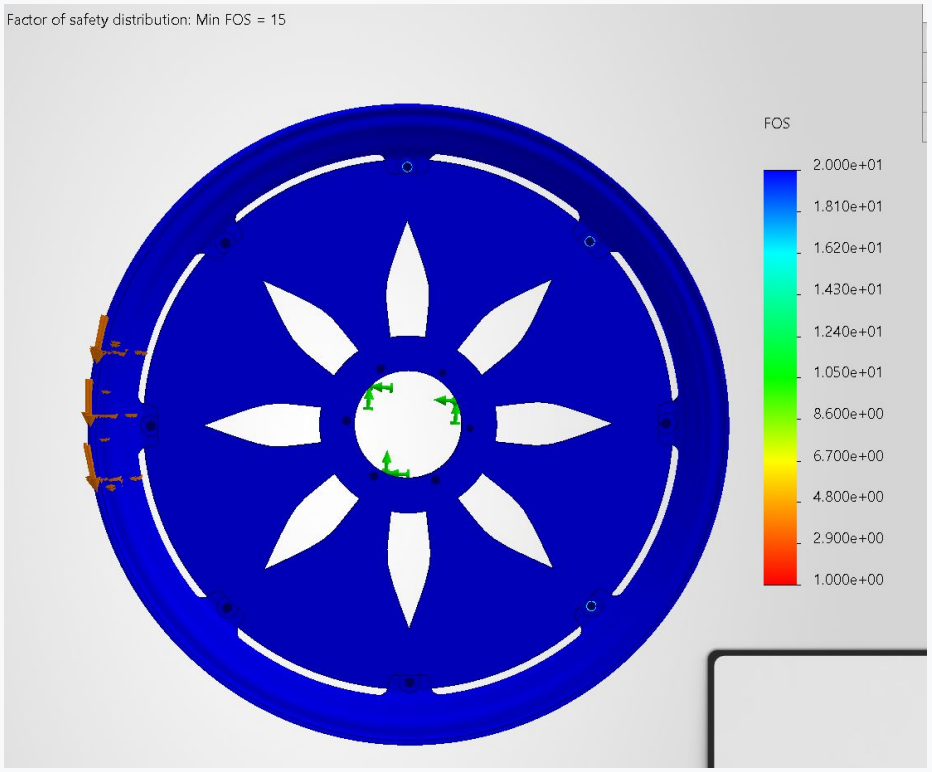


Results for bump force

- FOS: 15
- Yield Strength - 275 MPa
- Max displacement - 0.425 mm

First Design

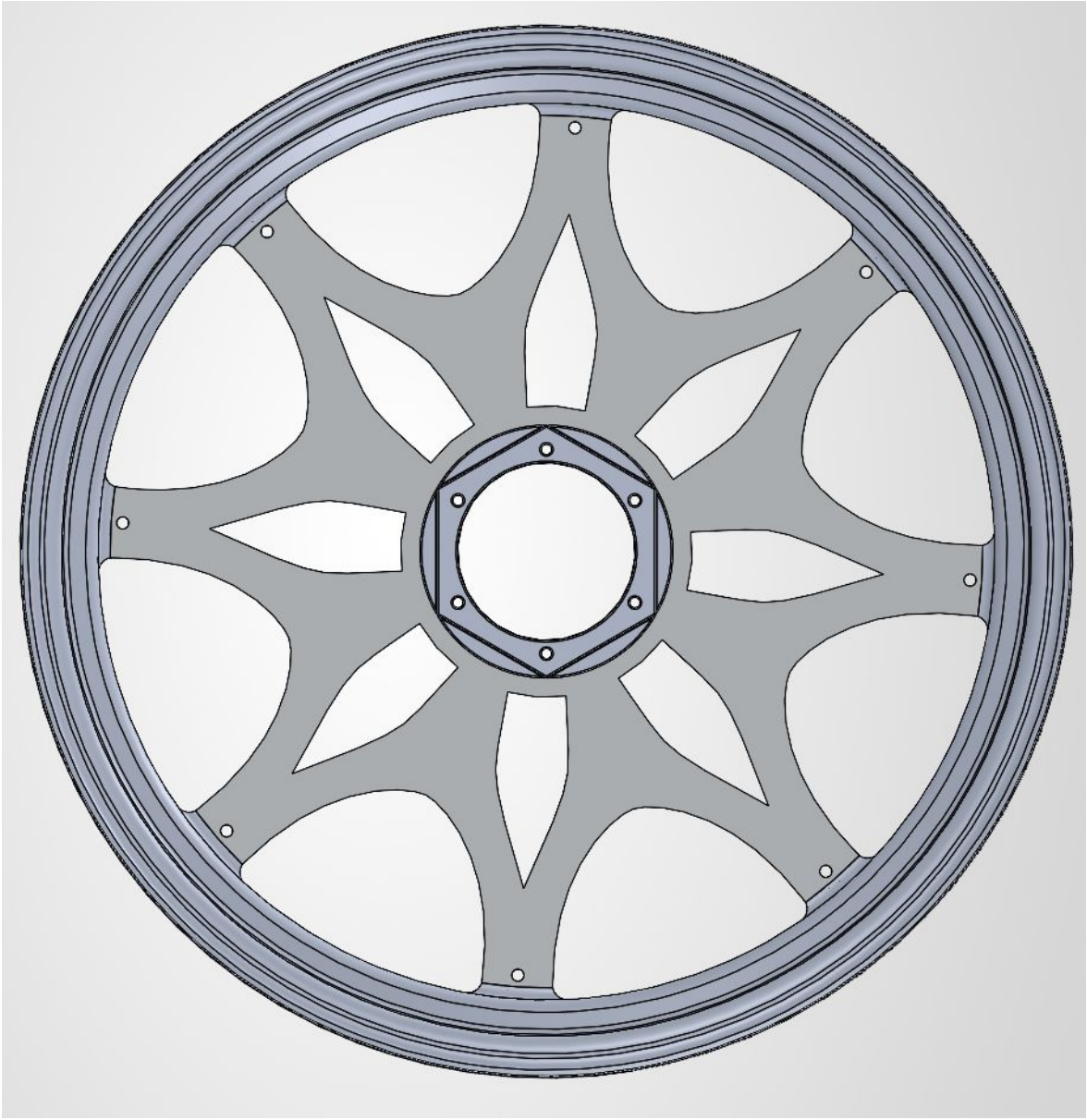




Decision:

- Keep load and material assumptions consistent
- Weight reduction (mass 1.127 kg)
- Lower the FOS number while keeping the same strength

Second Design



Material and mass

Item	Material or mass
Spoke body	6061-T6 (Aluminum Alloy)
Current spoke mass	581 g
Reference spokes	698 g

Reported comparison: 117 g lighter, approximately 16.8%

- Lightweight material, reducing mass with a high strength to weight ratio
- Easy manufacturability
- Can handle cyclical loading

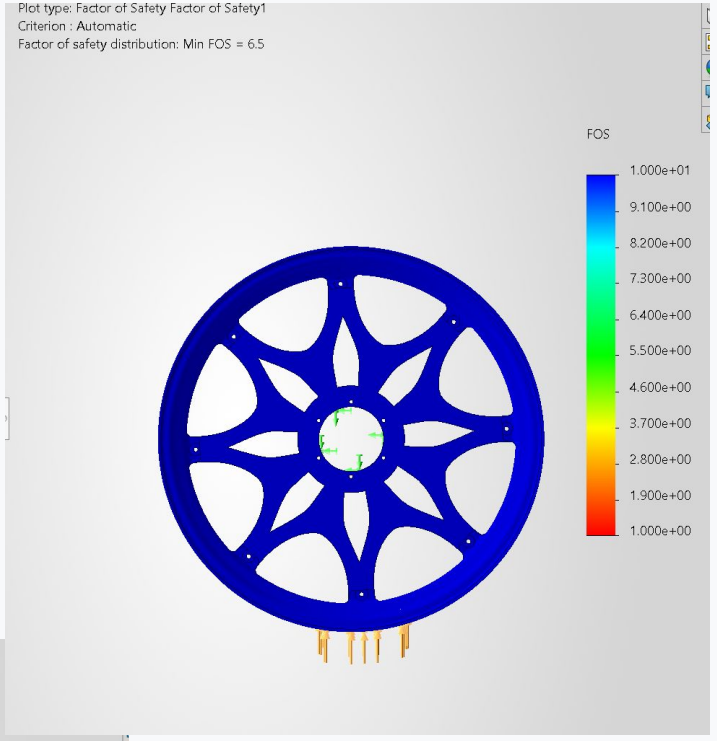
Simulation setup

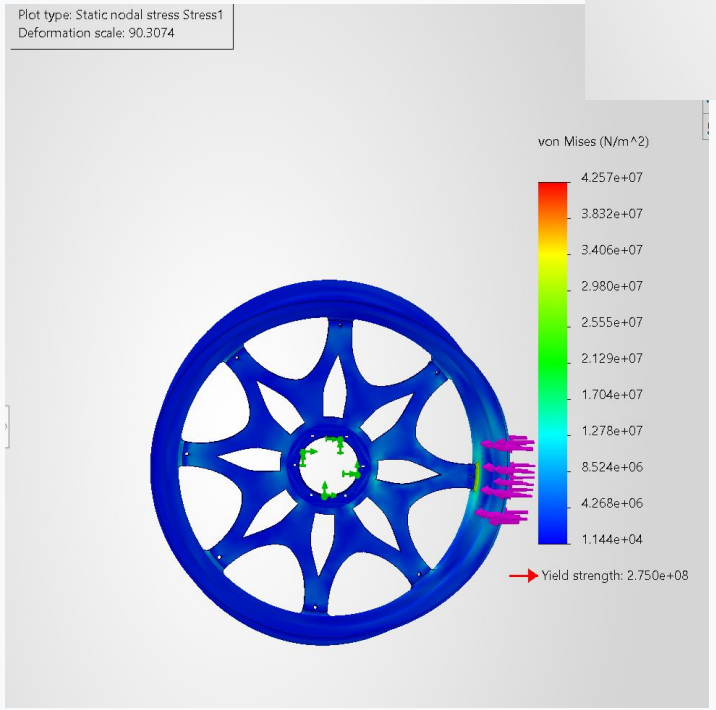
Fixtures

- Simulation is constraint at the centerlock
- Where the rim is going to be attached
- Isolate the system

Loads

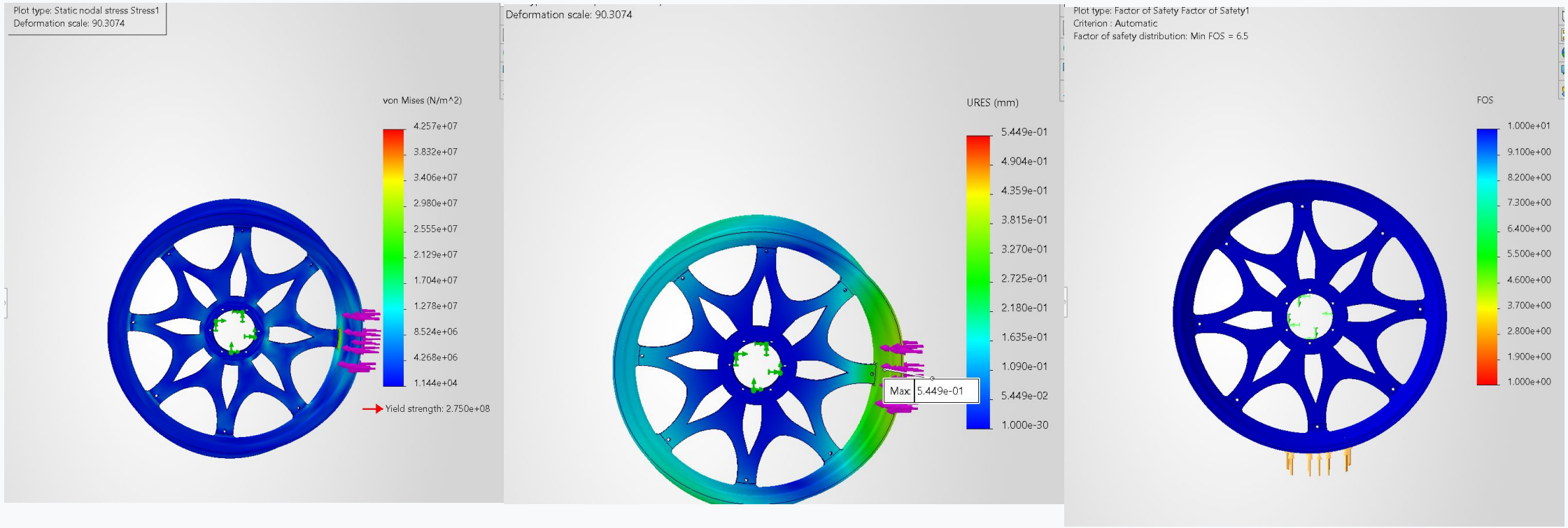
Bump: upward force on the rim patch.
Brake: torque about the rim patch





Bump results

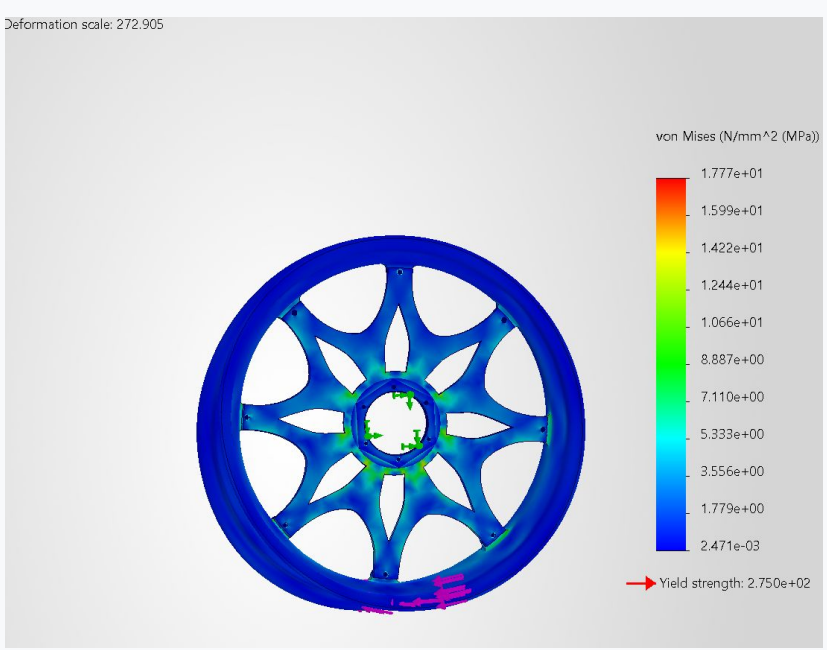
2g upward rim load; 275 MPA yield strength

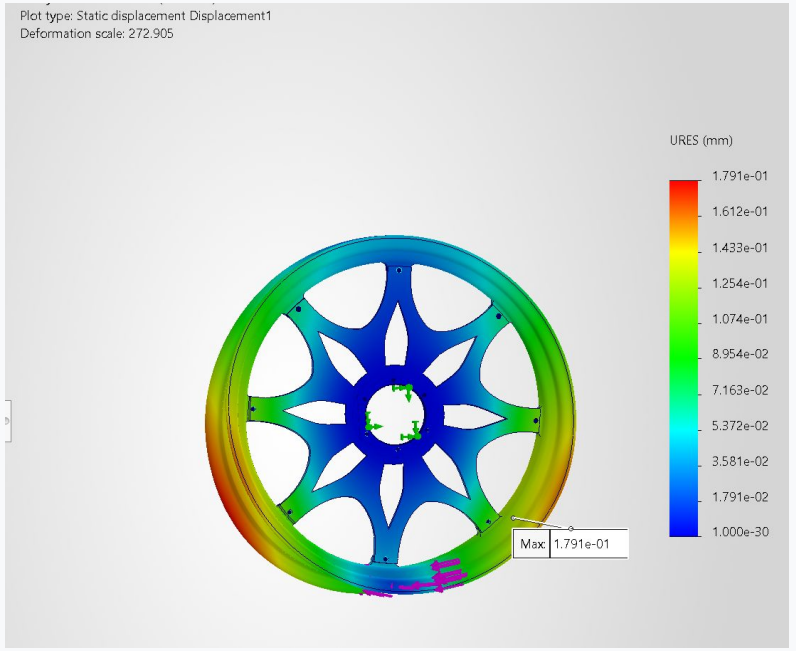


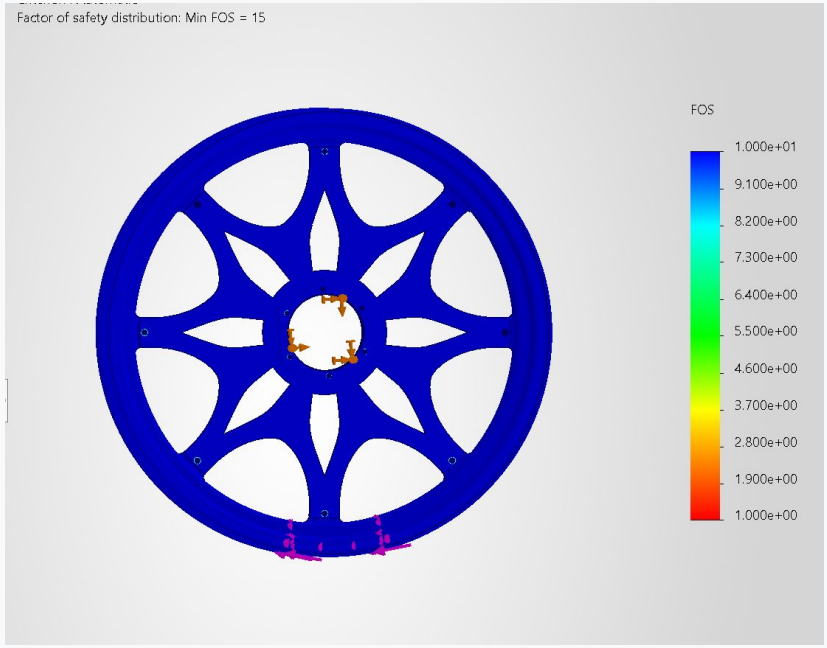
Max stress: 42.57 MPa **Min FOS: 6.5** **Max URES: 0.5449 mm**

Braking results

1g equivalent braking torque; 275 MPA yield strength







Max stress: 17.7 MPa Min FOS: 15.48 Max URES: 0.1791 mm

Manufacture and Desing Procedure

Decision:

- CNC machine for the spoke (best for aluminum alloys)
- Deburr edges and inspect thickness and hole locations, (engineering drawings, GD&T)

Materials:

- **Purchase aluminum stock and pins**
- **machine, inspect, assemble at work station**

What to improve?

- Experiment with different materials
- Could make the weight savings go down more
  - Current FOS is 15 for brake, can be lowered